

Towards Fair Energy Communities: Integrating Storage, Sharing and Pricing Strategies

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Abstract: This study employs a Mixed-Integer Linear Programming model to optimize energy transactions in an energy community. This research promotes energy sharing through grid and peer-to-peer transactions by incorporating energy storage systems. The article also integrates fairness metrics to evaluate economic equity. The case study examines households with varying equipment acquisition power, highlighting the impact of pricing structures on economic fairness. Analyzing optimal results, the study focused on a community of ten households, six of which are prosumers with PV and energy storage of different capacities. Results show that combining peer-to-peer transactions and high-capacity PV and storage can lead up to a 183% improvement in energy bills. On the other hand, members participating in energy sharing without PV and storage within the community only achieve ~13% improvement in their energy bill, clearly showing a disparity in cost savings linked to equipment capacity acquisition.

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Keywords: energy community, energy sharing, energy storage systems, fairness metrics, pricing strategies.

1. INTRODUCTION

The European Union (EU) is actively shaping its current energy strategy, emphasizing the central role of end-users in electricity markets (Lezama et al., 2021). Aligned with sustainability targets, the strategy addresses the need to reduce greenhouse gas emissions and increase renewable energy by 2030, adding complexity to the planning, operation, and protection of modern power systems (de Lima et al., 2023).

A key feature of this strategy is the promotion of energy communities (EC), particularly prosumer-dominated entities that support localized energy production (Faia et al., 2024). In these communities, local groups actively engage in generating, storing, distributing, and consuming renewable energy, empowering individuals in the transition to sustainable and decentralized energy systems (Gjorgievski et al., 2021). Despite several advantages that local generation brings for households, the declining appeal of selling generation surplus to the grid at feed-in tariffs (FiTs) has redirected them toward self-consumption. This shift is economically driven, outweighing the traditional practice of selling surplus generation back to the grid (Lezama et al., 2018).

In this context, peer-to-peer (P2P) energy sharing gives prosumers the autonomy to independently perform energy trading within their connected community (Tushar et al., 2019). This approach empowers end-users and facilitates flexible energy exchanges among peers within an EC. In P2P, prosumers exercise autonomy over critical energy-sharing parameters, deciding on the quantity and pricing of shared energy, as well as selecting specific community members for transactions (Zhang et al., 2022). It is important to note that, despite potential influences from a centralized controller or third party, prosumers retain the ultimate control over their

trading decisions (Tushar et al., 2021). By tapping into shared energy resources, including small-scale PV generation and energy storage, users aim to achieve lower financial costs, enhance self-sufficiency, and optimize the utilization of local resources, contributing to a more resilient, cost-effective, and community-centric landscape (Minuto & Lanzini, 2022).

In this work, we study key dynamics between energy sharing, pricing strategies, and fairness metrics. Employing a Mixed-Integer Linear Programming (MILP) model (Faia et al., 2021), optimal peer-to-peer transactions are achieved between community members. The integration of PV and energy storage systems of different capacities is a crucial aspect of this study, which further incorporates fairness metrics to evaluate economic equity between community members (Moradi Moghadam et al., 2022). The results indicate that all members can achieve economic benefits through enabling energy sharing. However, simultaneously, there exists a disparity in potential savings among members with different power acquisition capacities within the same community.

The paper is organized as follows. After the introduction, Section II presents the problem formulation, pricing mechanism and fairness metrics. Section III introduces the case study. Section IV presents the results and discussion to finally conclude the article in Section V.

2. PROBLEM FORMULATION

This section presents the mathematical formulation for optimizing surplus energy sharing within an EC that incorporates battery storage systems. The full problem formulation includes the optimization of energy-sharing, and two ex-ante pricing methods for calculating transaction costs. Additionally, two fairness metrics are applied to evaluate economic fairness, with the objective of achieving a balance between efficient energy sharing and economic equity.

2.1 Energy Sharing Strategy

This work optimally identifies decision variables related to grid imports, exports, peer transactions, battery charging and discharging, and energy bills by means of a MILP model proposed in (Dyngne et al., 2023; Faia et al., 2021).

The objective function searches for the minimization of energy bills considering multiple factors, including energy purchased from the grid, energy sold back to the grid, and the defined Time-of-Use and FiT prices for each operation:

$$\min_{P_{t,h}^{\text{buyG}}, P_{t,h}^{\text{sellG}}} \left(\sum_{h \in H} \sum_{t \in T} (C_{t,h}^R \cdot P_{t,h}^{\text{buyG}} - C_{t,h}^{\text{FiT}} \cdot P_{t,h}^{\text{sellG}}) \cdot \frac{1}{\Delta t} \right) \quad (1)$$

Here, t represents the period, h denotes the household, T is the total number of periods, and H is the total number of households; $C_{t,h}^R$ is the electricity purchase price from the retailer under a Time-of-Use tariff; $P_{t,h}^{\text{buyG}}$ is the electricity quantity bought from the retailer; $C_{t,h}^{\text{FiT}}$ is the selling price of electricity to the grid (i.e., the FiT); and $P_{t,h}^{\text{sellG}}$ represents the electricity quantity sold to the grid. The parameter Δt adjusts the tariff price to align with the optimization time intervals (for instance, 15-minute increments in this work).

The model is subject to different constraints. Equation (2) stipulates that the energy bought by one peer from another must equal the energy sold by the latter to the former:

$$P_{t,h,p}^{\text{buyEC}} = P_{t,p,h}^{\text{sellEC}} \quad \forall p \neq h \quad (2)$$

Here, $P_{t,h,p}^{\text{buyEC}}$ is the energy amount household h purchased from peer p , and $P_{t,p,h}^{\text{sellEC}}$ is the energy peer p sold to peer h .

Equation (3) quantifies the total energy purchased from all peers, excluding the house itself, while Equation (4) complements this by tabulating the total energy sold to peers, excluding transactions involving the house under consideration:

$$P_{t,h}^{\text{buyEC}} = \sum_{p \neq h} P_{t,h,p}^{\text{buyEC}}, \quad \forall t \in T, \forall h \in H \quad (3)$$

$$P_{t,h}^{\text{sellEC}} = \sum_{p \neq h} P_{t,p,h}^{\text{sellEC}}, \quad \forall t \in T, \forall h \in H \quad (4)$$

Here, $P_{t,h}^{\text{buyEC}}$ represents the total sum amount purchased in the EC market ($P_{t,h,p}^{\text{buyEC}}$), and $P_{t,h}^{\text{sellEC}}$ represents the total sum amount sold in the EC market ($P_{t,p,h}^{\text{sellEC}}$).

Equation (5) guarantees that the total energy acquired from the grid ($P_{t,h}^{\text{buyG}}$), energy procured from peers ($P_{t,h}^{\text{buyEC}}$), and energy generated from photovoltaic sources ($P_{t,h}^{\text{PV}}$), along with the discharging of household batteries ($P_{t,h}^{\text{dch}}$), align with the charging of batteries ($P_{t,h}^{\text{ch}}$), the demand ($D_{t,h}$), the energy sold to peers ($P_{t,h}^{\text{sellEC}}$) and the grid ($P_{t,h}^{\text{sellG}}$):

$$P_{t,h}^{\text{buyG}} + P_{t,h}^{\text{buyEC}} + P_{t,h}^{\text{PV}} + P_{t,h}^{\text{dch}} = P_{t,h}^{\text{ch}} + D_{t,h} + P_{t,h}^{\text{sellEC}} + P_{t,h}^{\text{sellG}}, \quad \forall t \in T, \forall h \in H \quad (5)$$

Binary conditions play a pivotal role in governing energy flow within the community, ensuring that specified limits are

observed to maintain a balanced and controlled energy exchange. The initial set of binary conditions in Equations (6) and (7) establishes that only one of the grid import or export, as well as one of the peer import or export, can be activated at any given time step:

$$x_{t,h}^{\text{buyGrid}} + x_{t,h}^{\text{sellGrid}} \leq 1, \quad \forall t \in T, \forall h \in H \quad (6)$$

$$x_{t,h}^{\text{buyEC}} + x_{t,h}^{\text{sellEC}} \leq 1, \quad \forall t \in T, \forall h \in H \quad (7)$$

Here, $x_{t,h}^{\text{buyGrid}}, x_{t,h}^{\text{sellGrid}} \in \{0,1\}$ represent the binary variables for grid import and export, respectively, while $x_{t,h}^{\text{buyEC}}, x_{t,h}^{\text{sellEC}} \in \{0,1\}$ denote the binary variables for EC import and export, respectively.

Equations (8-11) ensure that peer-to-peer energy exchanges remain within predefined thresholds, promoting a fair and sustainable energy-sharing ecosystem:

$$P_{t,h}^{\text{buyEC}} \leq P^{\text{maxbuyEC}} \cdot x_{t,h}^{\text{buyEC}} \quad (8)$$

$$P_{t,h}^{\text{sellEC}} \leq P^{\text{maxsellEC}} \cdot x_{t,h}^{\text{sellEC}} \quad (9)$$

$$P_{t,h}^{\text{buyG}} + P_{t,h}^{\text{buyEC}} \leq P^{\text{maxbuyG}} \cdot x_{t,h}^{\text{buyGrid}} \quad (10)$$

$$P_{t,h}^{\text{sellG}} + P_{t,h}^{\text{sellEC}} \leq P^{\text{maxsellG}} \cdot x_{t,h}^{\text{sellGrid}}, \quad \forall t \in T, \forall h \in H \quad (11)$$

Here, P^{maxbuyEC} and $P^{\text{maxsellEC}}$ set the maximum limits for purchasing and selling energy within the community, while P^{maxbuyG} and P^{maxsellG} determine the maximum allowable limits for purchasing and selling energy from and to the retailer.

2.2 Energy Storage Systems

Including energy storage systems in households within the energy community completes the representation of a diverse community. Equations (12) and (13) are the constraints for the charging and discharging capabilities of the batteries:

$$P_{t,h}^{\text{ch}} \leq P_h^{\text{max_ch}} \cdot x_{t,h}^{\text{ch}}, \quad \forall t \in T, \forall h \in H \quad (12)$$

$$P_{t,h}^{\text{dch}} \leq P_h^{\text{max_dch}} \cdot x_{t,h}^{\text{dch}}, \quad \forall t \in T, \forall h \in H \quad (13)$$

Here, $P_h^{\text{max_ch}}$ is the maximum power for battery charging, $P_h^{\text{max_dch}}$ is the maximum power for discharging, and $x_{t,h}^{\text{ch}}, x_{t,h}^{\text{dch}} \in \{0,1\}$ are binary variable linked to the charging and discharging state. Equation (14) avoid charge and discharge the battery in the same period:

$$x_{t,h}^{\text{ch}} + x_{t,h}^{\text{dch}} \leq 1, \quad \forall t \in T, \forall h \in H \quad (14)$$

Equation (15) is the state of charge of the batteries:

$$E_{t,h}^{\text{Bat}} = E_{t-1,h}^{\text{Bat}} + P_{t,h}^{\text{ch}} \cdot \eta_h^{\text{ch}} - \frac{P_{t,h}^{\text{dch}}}{\eta_h^{\text{dch}}}, \quad \forall t \in T, \forall h \in H \quad (15)$$

Here, $E_{t,h}^{\text{Bat}}$ is the battery's current state of charge, $E_{t-1,h}^{\text{Bat}}$ represents the state of charge in the previous period ($t-1$), $\eta_h^{\text{ch}} / \eta_h^{\text{dch}}$ is the charge/discharge efficiency.

Equations (16-18) are the constraints defining the upper and lower limits for the problem variables.

$$0 \leq P_{t,h}^{\text{dch}} \leq P_h^{\text{max_dcd}}, \quad (16)$$

$$0 \leq P_{t,h}^{\text{ch}} \leq P_h^{\text{max_ch}} \quad (17)$$

$$0 \leq E_{t,h}^{\text{Bat}} \leq E_{t,h}^{\text{max_Bat}}, \forall t \in T, \forall h \in H \quad (18)$$

Here, $E_{t,h}^{\text{max_Bat}}$ denotes the maximum capacity of the battery.

Notice that the optimization problem consider decision variables for the energy bought/sold from/to the grid ($P_{t,h}^{\text{buyG}}, P_{t,h}^{\text{sellG}}$), the energy bought/sold from/to the EC ($P_{t,h}^{\text{buyEC}}, P_{t,h}^{\text{sellEC}}$), and the energy exchanged between peers ($P_{t,h,p}^{\text{buyEC}}, P_{t,h,p}^{\text{sellEC}}$), for each household h at each time t . Solving the optimization problem also provides the battery state of charge ($E_{t,h}^{\text{Bat}}$) and the charge/discharge schedule ($P_{t,h}^{\text{ch}}, P_{t,h}^{\text{dch}}$) for each household h at each time t .

2.3 Pricing Mechanisms

In this work we assume that a community manager (CM) can apply two distinct pricing strategies.

In the first strategy, the local cost $C_{t,h}^{\text{EC}}$ is determined by the mid-market rate (MMR) pricing (Gjorgievski et al., 2022):

$$C_{t,h}^{\text{MMR}} = \frac{C_{t,h}^{\text{R}} + C_{t,h}^{\text{FiT}}}{2} \quad (21)$$

This pricing strategy is based on the average between the FiT and the grid price. With this pricing strategy, prosumers increase their revenue compared to the fixed FiT, while maintaining a competitive advantage over standard grid tariff.

In the second strategy, the supply-demand ratio (SDR) pricing mechanism is adopted (Dyngne et al., 2023). The underlying concept of this mechanism is to ensure that the price follows the surplus of available energy in the LEM relative to the demand. The supply-demand ratio is calculated by:

$$SDR_t = \frac{\sum_h (P_{t,h}^{\text{sellG}} + P_{t,h}^{\text{sellEC}})}{\sum_h (P_{t,h}^{\text{buyG}} + P_{t,h}^{\text{buyEC}})} \quad (22)$$

After this, the local price based on SDR is computed using:

$$C_{t,h}^{\text{SDR}} = (SDR_t \cdot C_{t,h}^{\text{FiT}} + (1 - SDR_t) \cdot C_{t,h}^{\text{R}}) \quad (23)$$

Equation (23) searches for a balance between the surplus measure (SDR_t) and upper and lower boundaries in prices (i.e., the buying price $C_{t,h}^{\text{R}}$ and selling price $C_{t,h}^{\text{FiT}}$). Therefore, in a balanced market (i.e., $SDR_t = 0$), the local price aligns with the upper bound defined by the grid cost ($C_{t,h}^{\text{R}}$). Conversely, in the presence of surplus of energy (i.e., $SDR_t \geq 1$), the local price corresponds to the lower bound represented by the FiT ($C_{t,h}^{\text{FiT}}$).

2.4 Fairness metrics

Fairness metrics play a pivotal role in assessing equitable benefits within energy communities, especially considering battery storage systems (Lüth et al., 2018). These metrics critically evaluate economic fairness by examining aspects such as peer-to-peer transactions, pricing strategies, and overall cost reductions.

We employ Quality of Experience fairness (QoE) and minimum-maximum fairness (mMF) (Dyngne et al., 2023) as metrics to assess fairness of the pricing mechanisms. To compute these metrics, the overall Energy Bill (EB_h) under the selected pricing mechanisms (i.e., C_t^{EC}) is defined by:

$$EB_h = (C_{t,h}^{\text{R}} \cdot P_{t,h}^{\text{buyG}} - C_{t,h}^{\text{FiT}} \cdot P_{t,h}^{\text{sellG}} + C_t^{\text{EC}} \cdot (P_{t,h}^{\text{buyEC}} - P_{t,h}^{\text{sellEC}})) \cdot \frac{1}{\Delta t} \quad (24)$$

This expression encapsulates the economic impact of the pricing mechanisms on the energy bill, considering grid transactions, feed-in tariffs, and peer-to-peer exchanges.

The Quality of Experience (QoE) fairness metric evaluates the diversity in costs among participants within the EC:

$$QoE_t = 1 - \frac{\sigma_t}{\sigma_t^{\text{max}}}, \forall t \in T, \forall h \in H \quad (25)$$

$$\sigma_t = \frac{EB_h}{D_{t,h} - P_{t,h}^{\text{PV}}} \quad (26)$$

$$\sigma_t^{\text{max}} = C_t^{\text{R}} - C_t^{\text{FiT}} \quad (27)$$

where σ_t is the standard deviation of perceived prices, reflecting the degree of price variability relative to the energy bill (EB_h) divided by the disparity between energy demand ($D_{t,h}$) and photovoltaic power ($P_{t,h}^{\text{PV}}$) (i.e., Eq. (26)). The maximum deviation, σ_t^{max} , is determined by the subtraction of the retailer price and related taxes (C_t^{R}) from the Feed-in Tariff (C_t^{FiT}) (i.e., Eq. (27)). A QoE_t value of 1 means uniform perceived prices among participants, indicating a more equitable distribution of economic benefits across households.

The Minimum-Maximum Fairness (mMF) is computed as the ratio of the minimum to the maximum energy bill:

$$mMF_t = \frac{\min_h(EB_h)}{\max_h(EB_h)}, \forall t \in T, \forall h \in H \quad (28)$$

A higher mMF value, approaching 1, represents minimal disparity among participants' energy bills.

In the context of this article, these metrics contribute to assessing the equality of PV and storage systems, as those captures the economic impact of pricing mechanisms on different households, including those with varied capacities.

3. CASE STUDY

The case study comprises ten active members situated in a suburban area, fostering the exchange of surplus energy. Among these community members, six are prosumers equipped with PV panels and varying capacities of battery storage. The analysis scrutinizes a single day at 15-minute intervals, resulting in 96 data points of generation data taken from GECAD Building N in Porto, Portugal (Goncalves et al., 2022). To simulate diverse income levels within the community, profiles for six prosumers are derived from a common base profile, introducing slight random variations. Households 1 and 2 showcase low PV, peaking at around 4 kW, while households 3 and 4 are categorized as mid PV with approximately 8 kW of peak power. Finally, households 5 and 6 are categorized as high PV (e.g., having additional PV panels) achieving a peak power near to 12 kW.

Similar to the PV classifications for the same six households, battery systems also span three capacity levels—low, mid, and high. Detailed configurations are presented in Table 1, with diverse models of storage units (Faia et al., 2023). For instance, households 1 and 2, classified as low, have batteries with a maximum capacity of 6.4–6.6 kWh, while households 3 and 4, designated as mid, are equipped with 9.3–9.6 kWh batteries. Households 5 and 6, classified as high, have batteries with a capacity of 12.9 kWh and 17.2 kWh respectively. All batteries have efficiencies at or above 95%.

Table 1. Household Battery Storage Specifications. Taken from (Faia et al., 2023).

Household	Capacity (kWh)	Charge (kW)	Discharge (kW)	Efficiency (%)	Initial Final SoC (kWh)
1–Low	6.4	2	2	0.95	3.3
2–Low	6.6	3.5	3.5	0.945	3.3
3–Mid	9.3	5	5	0.95	4.8
4–Mid	9.6	5	5	0.975	4.8
5–High	12.9	6.5	6.5	0.96	6.45
6–High	17.2	8.5	8.5	0.975	8.6

Households 7 to 10 do not have battery systems.

Finally, households 7 to 10 function as energy consumers, with consumption profiles from diverse European households (Goncalves et al., 2022). Contracted power values are aligned with fixed powers in Portugal with maximum consumption levels for end-users (e.g., 3.45, 6.9, and 10.35 kW).

To establish a benchmark for comparing energy bills, we computed the Business as Usual (BAU) energy bill for each member. This process involved utilizing individual demand and PV generation data without any energy sharing mechanism. We applied the retailer's triple tariff (from www.suelectricidade.pt) for energy imported from the grid and the FiT (0.045 EUR/kWh) for surplus energy exported to the grid. The BAU case provides a baseline for assessing the costs and profits experienced by end-users without any optimization or energy sharing mechanism.

4. RESULTS AND DISCUSSION

The experimentation was carried out using the Python programming language within the Spyder environment in a computing system featuring an Intel(R) Core (TM) i5-8250U CPU @ 1.60GHz processor, 20 GB of RAM, and operating on the Windows 11 platform. The MILP model was solved by the Coin-or branch and cut (CBC) solver in Python MIP library.

4.1 Energy sharing transactions

Figure 1 shows the sellers along the xx-axis, buyers along the yy-axis, and the corresponding transaction volume during the entire day along the zz-axis. Evidently, successful energy sharing occurs among households, with varying degrees of participation. Notably, the most substantial volume of energy sharing occurs between households 2 and 6, with a total of 42.41 kWh, followed by the transactions between household 1

and 6 (totaling 15.79 kWh), and between households 2 and 5 (with a volume of 14.88 kWh). On average, each prosumer in the P2P market transacted approximately 5.88 kWh during the day. It is worth mentioning that households 7 to 10 do not sell energy in the P2P market due to the absence of PV and battery systems in these households.

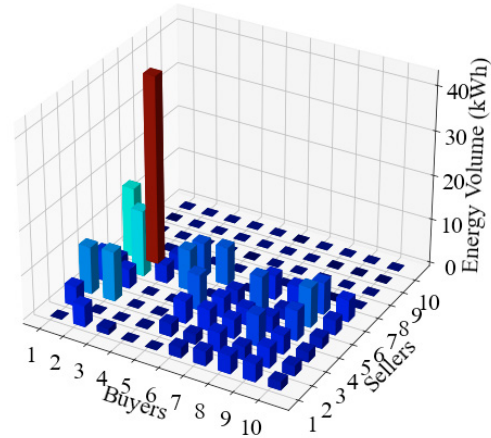


Figure 1. Energy sharing transaction volume between peers.

4.2 Battery Storage System

Figure 2 depicts the battery capacity throughout different periods of the day for households equipped with energy storage of varying capacities. The charge and discharge cycles underscore the essential contribution of energy storage to daily household operations. Notice that households 5 and 6 (with highlighted color in the figure), categorized as High PV and battery, achieve full battery charge (with 6.5 kWh and 8.5 kWh, respectively) around period 40, to later discharge the battery in night periods (i.e., after period 60). The highest storage levels are observed around noon, coinciding with the peak solar exposure.

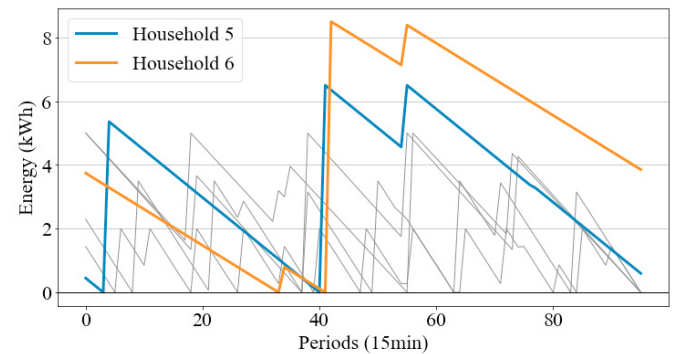


Figure 2. Usage of batteries during the day. Households 5 and 6 are highlighted with blue and orange colors respectively for a better appreciation of their profiles.

Figure 3 provides a detailed overview of the charging power at each 15-minute interval throughout the day. All prosumers make use of their battery system during the day showcasing active participation in all periods. This makes evident that households exploit the triple tariff from the retailer, buying energy in periods of low prices to use that energy in later times.

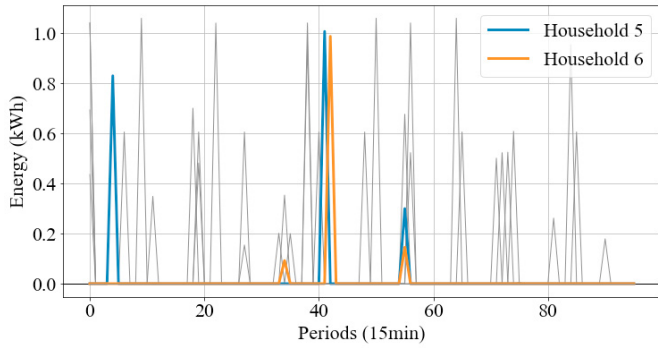


Figure 3. Battery charge profiles. Households 5 and 6 are highlighted with blue and orange colors respectively for a better appreciation of their profiles.

4.3 Pricing and bill comparison

Table 2 presents the average energy cost (in EURs) across segments of households and pricing strategies. The BAU column serves as the benchmark cost. The MMR and SDR columns illustrate the costs under the MMR and SDR pricing strategies, respectively. Columns MMR_{imp} and SDR_{imp} present the percentage improvements regarding the BAU scenario (with negative values denoting cost savings). Cost savings are achieved for all segments of households (i.e., with low-mid-high PV-battery), demonstrating the viability of energy sharing. However, notice that the higher the capacity of the PV and battery system, the higher the savings obtained by the members of the community. For instance, high PV-battery households obtain on average an improvement of 183% with the MMR strategy, and 150% with the SDR, w.r.t. the BAU scenario. These improvements are much more than the savings obtained by other segments of households (for instance, only 13% and 19% of savings for households without PV and battery systems). This result highlights the effectiveness in cost reduction using energy sharing and pricing strategies, but at the same time shows a large disparity of cost savings depending on the capacity for equipment acquisition.

Table 2. Average Energy Cost Comparison Across Scenarios and Pricing Strategies

Households	BAU (EUR)	MMR (EUR)	MMR_{imp} (%)	SDR (EUR)	SDR_{imp} (%)
1 and 2 (Low PV Battery)	5.95	2.45	58.82	2.01	66.22
3 and 4 (Mid PV Battery)	2.12	-0.72	133.96	-0.51	124.06
5 and 6 (High PV Battery)	2.12	-1.78	183.96	-1.06	150
7 to 10 (No PV Battery)	3.72	3.23	13.17	2.98	19.89

Figure 4 shows graphically the differences in cost savings depending on the capacity of PV-battery system. While savings are achieved for all groups of households w.r.t. the BAU scenario, it is quite evident that the savings achieved by community members are tightly related to the capacity for equipment acquisition.

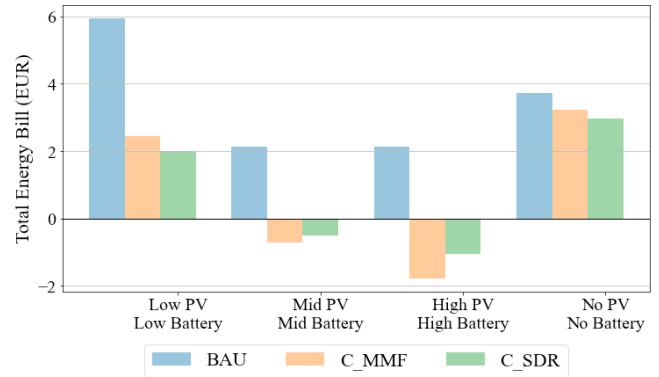


Figure 4. Cost comparison for different scenarios under various pricing strategies.

4.4 Fairness Analysis

Finally, Figure 5 shows the evolution of the fairness metrics (See Sect. 2.4) over the different pricing schemes. Figure 5(a) shows that the QoE Index improves with both pricing schemes (green and orange lines in the figure) compared to the grid triple tariff (blue line). Notice also that the MMR pricing scheme (orange line) shows better QoE Index in periods where PV is not available, while SDR (green line) presents better performance when PV is available. On the other hand, notice in Figure 5(b) that the mMF metric is null in periods with absence of PV generation, making evident that this metric requires energy sharing to provide a sense of fairness. While the values are different to those of QoE Index, the trend in PV generation periods is the same, resulting in a better performance of the SDR pricing scheme (green line) over the MMR (orange line) in those periods.

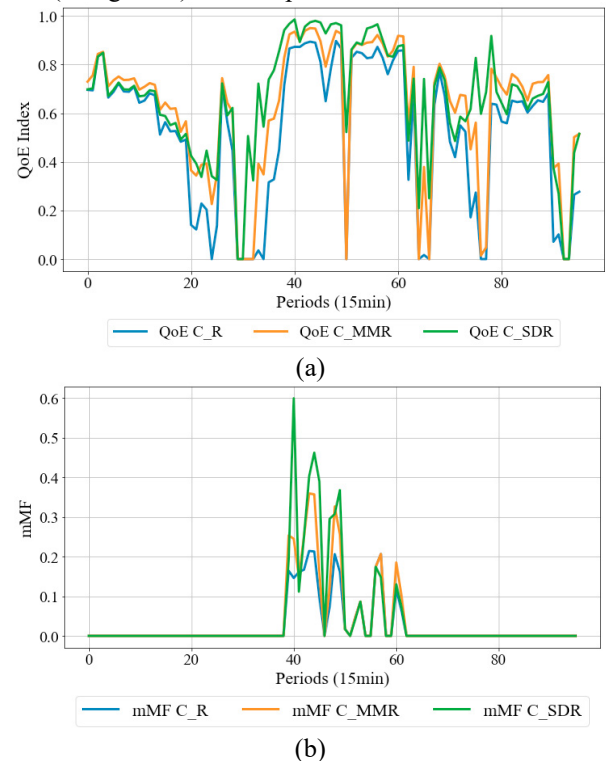


Figure 5. Cost comparison for different scenarios under various pricing strategies.

5. CONCLUSIONS

In this article, we explored the dynamics of energy-sharing transactions in an energy community (EC), assessing the impact of different pricing schemes on photovoltaic (PV) and battery storage systems. The results demonstrate cost savings for all community members, emphasizing the efficacy of both MMR and SDR pricing strategies. However, a significant disparity in savings was observed, directly tied to the capacity of PV and battery systems in some households, economically benefiting those with higher acquisition capacity. The linkage between equipment capacity and cost reduction plays a key role in the search for success in energy sharing and fairness in energy communities. Furthermore, our fairness analysis introduced the QoE Index and mMF metrics, demonstrating their sensitivity to pricing schemes and the availability of PV as well. The nuanced performance of these metrics underscores the context-dependent nature of fairness assessments. Overall, these findings emphasize the need for tailored solutions considering both economic and social dimensions, recognizing the dynamic nature of fairness metrics in diverse energy-sharing contexts. Future research must further analyze social aspects in EC, focusing not only on the monetary advantages of member participants but also on the interactions of households with different income levels.

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